NREFT, Lattice & Form Factors

Ajay S. Sakthivasan[1], Akaki Rusetsky[1]

[1]HISKP & BCTP, University of Bonn

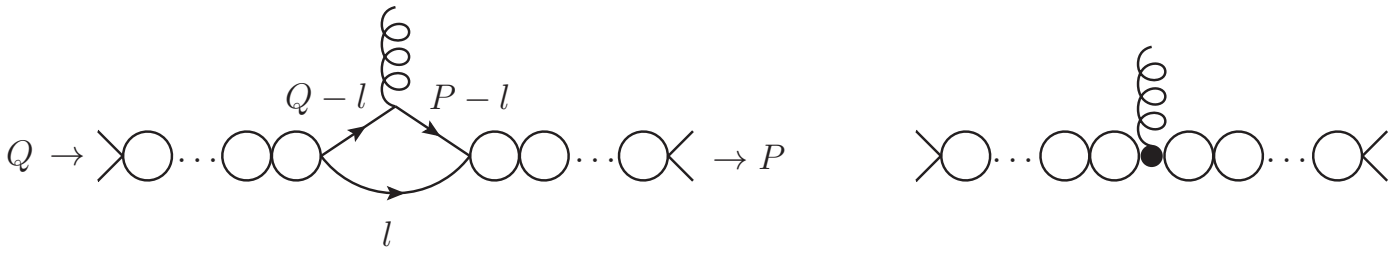
September 23, 2025

Form Factor Fundamentals

- Objective: the calculation of the ρ-meson electromagnetic form factor.
- The calculation of the form factors of unstable states on the lattice pose significant challenges.
- Given a quantum field with the quantum numbers of the ρ-meson obeying relevant constraints:
  - $\rightsquigarrow$ The two-point function develops a simple pole on the second Riemann sheet at $s = s_R$.
  - $\rightsquigarrow$ The three-point function involving the electromagnetic current develops a double pole at $s = s' = s_R$.
  - $\rightsquigarrow$ The extraction of the EM form factor corresponds to calculating the residue of the three-point function.

The Problems with the Form Factor

- The contributions can be split into a triangle diagram and the contact term:



(a) (b)

- The triangle diagram does not exhibit a regular dependence on the box size L. (*Bernard et al., 2012*)
- To overcome this, we make use of the external field method. (*Lozano et al., 2022*).
  - ⇝ The system is placed in a periodic external field, which is coupled to a given current.
  - ⇝ The modified Lüscher approach lets one extract relevant physical results.

The Form Factor in the NREFT Framework

- We work with an NREFT Lagrangian that describes the elastic $\pi^+\pi^0$ scattering with a dipion field Ψ that models the ρ-meson:

$$\mathcal{L} = \overbrace{\sum_{a=+,0} \phi_a^\dagger 2W(i(v\cdot\partial) - W)\phi_a}^{\text{Pion Lagrangian}} - \overbrace{\Psi_\mu^\dagger \Psi^\mu}^{\text{Rho Lagrangian}} \quad .$$

$$W = \sqrt{M^2 + \partial^2 - (v\cdot\partial)^2} \qquad \text{quantisation axis}$$

$$\Psi^\mu = \phi_+ \overset{\leftrightarrow}{W}{}^\mu f(\overset{\leftrightarrow}{W}{}^2)\phi_0,$$
$$W^\mu = v^\mu W + i(\partial^\mu - v^\mu (v\cdot\partial)),$$
$$f(\overset{\leftrightarrow}{W}{}^2) = f_0 + f_1 \overset{\leftrightarrow}{W}{}^2 + \ldots.$$

- f is related to the $\pi\pi$ scattering phase shift in the p-wave:

$$(q_0 \cot\delta_1(q_0))^{-1} = -\frac{q_0^2 f^2(q_0^2)}{24\pi\sqrt{P^2}}$$

Inclusion of Electromagnetic Interaction

- Gauging is non-trivial due to the presence of non-local operators.
- We consider the operators corresponding to the relevant invariant structures:

$$\mathcal{L} \supset \sum_{i=1}^{3} g_i e \mathscr{O}_i,$$

where

$$\mathscr{O}_1 = \Psi_\mu^\dagger \{\partial^\beta F_{\alpha\beta}, W_\alpha\} g^{\mu\nu} \Psi_\nu,$$
$$\mathscr{O}_2 = \Psi_\mu^\dagger i F^{\mu\nu} \Psi_\nu,$$
$$\mathscr{O}_3 = \Psi_\mu^\dagger [W^\mu, [W^\nu, \{\partial^\beta F_{\alpha\beta}, W_\alpha\}]] \Psi_\nu.$$

- The matrix element of the electromagnetic current between the two-pion states can be evaluated, and it is parametrised in terms of the couplings g_1, g_2 and g_3.

Matching of the Couplings to ChPT Amplitude

- The couplings g_1, g_2 and g_3 can be matched to $\mathcal{O}(p^4)$ ChPT LEC's, which yields:

$$eg_2 f^2(0) = \frac{2e(2l_1^r - l_2^r)}{F_\pi^4} + \text{loops},$$

$$eg_1 f^2(0) = eg_3 f^2(0) = 0 + \text{loops}.$$

- We make use of Schenk parametrisation for the phase shift, which yields

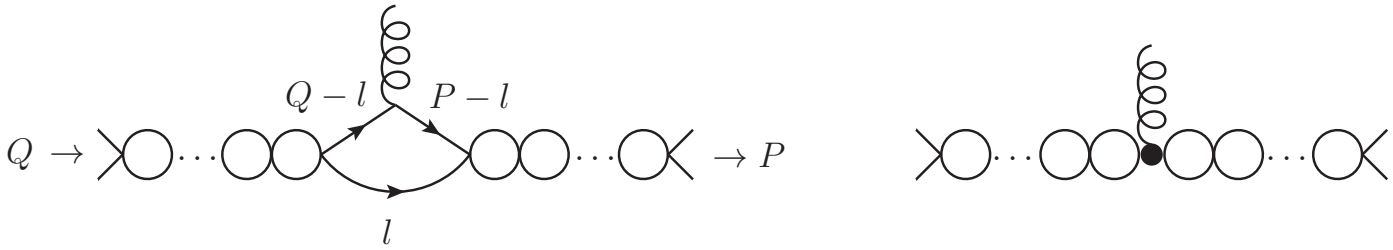
$$g_2 = -\frac{M_\pi^2(2l_1^r - l_2^r)}{24\pi F_\pi^4 a_1^1} + \text{loops} = 2.8 \cdot 10^{-2} M_\pi^{-2},$$

$$g_1 = g_3 = 0 + \text{loops}.$$

- This matching gives a numerical estimate of the couplings—which are regular and can be extracted from lattice.

The Form Factor in the Infinite Volume

- As mentioned before, the contributions to the form factor are from a triangle diagram and the contact term.



(a) (b)

- The contributions can now be calculated in the NREFT and related to the invariant form factor parameters.

Lüscher Equation in the External Field

- Placing the system in a periodic electromagnetic field on a cubic lattice allows one to extract the invariant form factors.
- For an unstable particle, the energy levels are determined by the modified Lüscher approach with the quantisation condition:

$$\det(\Pi^{-1}_{\mu\nu} - D^{0}_{\mu\nu}) = 0$$

"self-energy" the two-point function of the interpolating field

Summary

- As mentioned before, the couplings g_1, g_2 and g_3 can be evaluated by projecting on to the relevant irreps, which can be used to evaluate the infinite volume invariant form factors.
- One can also, obtain numerical estimates of g_1, g_2 and g_3 starting from infinite volume form factors.
- The Modified Lüscher equation in a background electromagnetic field allows one to extract the EM form factor of the ρ-meson on the lattice.